Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building and testing cross-platform applications using **C++, Python, Java**, object-oriented design, concurrent processing, and CI/CD, with hands-on simulated **SCPI waveform acquisition and signal measurement**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level isolation and normalized schemas, supporting 500+ users while reducing indexed-query latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous orchestration with explicit state transitions and failure isolation, processing 1K+ jobs while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Designed a concurrent batch-processing pipeline using multithreading and secure PDF, DOCX, and ZIP ingestion, processing 1K+ files per batch through AWS RDS and REST APIs with fault-tolerant execution.
- Implemented production Python services handling 10K+ daily requests for 150+ instructors, using CI/CD-backed workflows to reduce feedback turnaround 60% while preserving stable application performance during peak load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built a Python simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% while rigorously testing data transformations and model behavior across repeatable autonomous-driving benchmark evaluation scenarios.
- Architected a spatial-temporal encoder combining map geometry and trajectory context through multi-head attention, improving closed-loop stability 15% through systematic simulation, regression testing, detailed failure analysis, and debugging.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Engineered cross-platform microservices with automated VM builds and 20+ robust unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments.
- Optimized object-oriented API integrations and AWS Glue and S3 data flows, improving response times 30%, pipeline efficiency 68%, and long-term maintainability through reusable Factory and Singleton patterns.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Developed a MERN resource-management platform using WebSockets, REST APIs, and eight modular services, synchronizing 2M+ records across distributed locations while reducing production release-cycle code-review turnaround by 25%.
- Automated Docker-based application deployments through Jenkins CI/CD and integrated test validation, cutting deployment time 50% while achieving 95%+ successful build validation across repeated production software release runs.

Skills

Languages: C++, Python, Java, C#, JavaScript, TypeScript, SQL, Bash
Systems & Measurement: OOP, SCPI, Waveform Analysis, Signal Measurements, Concurrency, Multithreading, Operating Systems, Algorithms, Data Structures, REST APIs
Platforms: Windows, Linux, AWS, Azure, Docker, Kubernetes, PostgreSQL, SQL Server, MySQL, S3, Blob Storage
Quality: Git, CI/CD, Jenkins, GitHub Actions, PyTest, Mockito, Unit Testing, Integration Testing, Regression Testing, Debugging, Code Reviews, Agile

Projects

ScopeLink | Python, SCPI, Signal Analysis, Instrument Simulation
- Built and tested a simulated SCPI oscilloscope client for channel acquisition, response validation, RMS, peak-to-peak, frequency, and rise-time measurements, using interpolated zero crossings to measure 100 Hz accurately.

CPU-scheduling Simulator Live | C++, React.js, Operating Systems
- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling algorithms with interactive visualization and dynamic waiting-time and turnaround-time calculations across 1K+ repeatable simulated process combinations and edge cases.

Chat2SQL Code | Java, TypeScript, SQL, Docker
- Built a cross-platform database workspace supporting 10+ engines, combining Java and TypeScript services with Dockerized workflows for schema exploration, SQL generation, query authoring, and structured data management.